

CB-Signal: Extracting a Monetary-Policy Tone Signal from Central Bank Communications

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Abstract

We build an end-to-end pipeline that scrapes European Central Bank (ECB), Federal Reserve (Fed) and Bank of England (BoE) communications, scores their hawkish/dovish tone with three independent methods (a curated phrase dictionary, a pretrained financial-sentiment transformer, and a lightweight fine-tuned sentence-embedding classifier), aggregates tone into a weekly index, and backtests a resulting trading signal against 2Y/10Y government bond yields. On 3,968 real documents spanning 1997–2026, the baseline dictionary signal shows no robust full-sample edge (Sharpe ratios from -1.6 to $+0.07$ across legs), but sub-period analysis reveals the signal performs materially better during the 2022–2023 hiking cycle (Sharpe up to 1.8) than in other regimes. On a common window comparing all three sentiment tiers, the simple dictionary outperforms both the pretrained and fine-tuned transformer tiers — a caution against assuming more sophisticated models automatically win on a narrow, low-data task. We also document and fix a look-ahead issue in the original regime-labelling procedure, and report where a walk-forward refit changes the resulting labels. We report all of this plainly, including the negative full-sample result, rather than presenting only the sub-period in which the signal happens to work.

1 Motivation

Central bank communications are a primary channel through which monetary policy expectations are formed. A large academic literature [Apel and Blix Grimaldi, 2012, Hansen and McMahon, 2016, Bennani and Neuenkirch, 2017] argues that the *tone* of this language — hawkish (tightening-oriented) versus dovish (easing-oriented) — carries information beyond the policy rate decision itself. This project asks a narrower, falsifiable question: does a systematic, text-derived tone signal, built with openly documented methods and tested on real market data, produce a backtestable edge on government bond yields once realistic transaction costs are applied?

2 Data

2.1 Communications corpus

We scrape three sources directly from the issuing institutions’ own websites (no third-party aggregator):

- **ECB:** 1,871 speeches (1997–2026), discovered via the ECB’s AddSearch-powered site search (the previously published bulk CSV feed was discontinued following a site redesign) and parsed from each speech’s canonical HTML page.

- **Fed**: 1,649 documents — 167 FOMC statements, 150 FOMC minutes, and 1,332 speeches (2006–2026) — discovered via the Fed’s historical and current meeting calendars and speech archives.
- **BoE**: 448 documents — 371 speeches and 77 Monetary Policy Summary/minutes releases (2015–2026) — discovered via BoE’s sitemap feed, after the browsable listing pages moved to a JavaScript-rendered search widget.

Table 1 summarises coverage. All ingestion code retries transiently-failing requests, caches raw HTML on disk, and is designed to degrade gracefully (skip, don’t crash) when an individual page is unreachable.

Source	Documents	Event types	Date range
ECB	1,871	speech	1997–2026
Fed	1,649	statement, minutes, speech	2006–2026
BoE	448	speech, minutes	2015–2026
Total	3,968		1997–2026

Table 1: Real communications corpus used in this study.

2.2 Market data

Weekly 2-year and 10-year US Treasury yields (FRED series DGS2, DGS10), German and UK long-term government yields, and the VIX are pulled directly from the FRED API (39,665 observations, 1956–2026).

3 Methodology

3.1 Sentiment scoring tiers

We score every document with three independent methods so that the dictionary baseline’s results can be cross-checked rather than taken on faith:

1. **Dictionary** (baseline): a curated hawkish/dovish phrase lexicon drawn from Apel and Blix Grimaldi [2012], Hansen and McMahon [2016] and Bennani and Neuenkirch [2017], plus a Loughran–McDonald [Loughran and McDonald, 2011] uncertainty subset, matched with negation handling over a 3-token window. Net tone is (hawkish hits – dovish hits)/total hits.
2. **Pretrained transformer**: ProsusAI/FinBERT [Araci, 2019], a BERT model fine-tuned on financial news sentiment (positive/negative/neutral), with positive/negative remapped to dovish/hawkish. Long documents are chunked into overlapping 512-token windows and the resulting per-chunk probabilities averaged. (The FOMC-specific Shah et al. [2023] RoBERTa model would be a closer match to this task, but its HuggingFace repository is access-gated; FinBERT is used as the publicly reproducible alternative.)
3. **Fine-tuned tier**: sentence embeddings from `all-MiniLM-L6-v2` [Reimers and Gurevych, 2019] with a multinomial logistic-regression head, trained on a small (482-example) hawkish/dovish/neutral set. *This set is not an independently human-annotated research dataset*

— it is built by dropping each phrase from the dictionary lexicon into several carrier sentences, plus a hand-written list of genuinely neutral central-bank boilerplate sentences for the neutral class. We report it as a lightweight, transparently-constructed classifier, not as a validated gold-standard benchmark.

3.2 Signal construction

Document-level tone is averaged per source per week, then averaged across sources into a single weekly hawkish index. An exponential moving average (12-week half-life) serves as the baseline; the surprise (index $-$ EMA) is expanding-window z-scored (so, by construction, only information available up to that week is used — no look-ahead) and clipped to $[-3, 3]$. The tradable signal is a 4-week rolling average of a 50/50 blend of the z-scored level and the z-scored surprise, applied with full weight to the 2-year leg and 60% weight to the 10-year leg.

3.3 Regime labelling: a look-ahead issue, found and fixed

A 3-state Gaussian HMM on the level and week-over-week change of the hawkish index labels each week dovish/neutral/hawkish for descriptive and diagnostic purposes. *This regime label does not feed the tradable signal above* — the signal is built only from the already point-in-time EMA/surprise construction. The original implementation fit the HMM once on the entire sample, meaning early-sample regime labels were informed by the model’s fit to *later* data: a look-ahead issue for a descriptive label, even though it has no effect on realised backtest P&L. We implemented a walk-forward version that refits the HMM yearly on an expanding window and labels only the following year out-of-sample. The walk-forward and in-sample labels disagree on 45% of weeks, showing the original full-sample fit was not a robust description of the regime structure, and this out-of-sample version should be preferred whenever the regime label itself is used for anything beyond the illustrative plot included here.

3.4 BERTopic thematic decomposition

As a complementary descriptive layer, we fit BERTopic [Grootendorst, 2022] on sentence embeddings of the full corpus to recover the dominant themes discussed across communications (inflation, labour markets, financial stability, etc.) and their prevalence over time. This is reported alongside the tone signal as context, not as an input to it.

3.5 Backtest

Weekly yield changes are converted to approximate bond returns via a constant modified-duration approximation ($D_{2Y} = 1.9$, $D_{10Y} = 8.5$). Positions are $-$ sign/magnitude of the (lagged, $T + 1$) signal, clipped to $[-1, +1]$, with 20bps round-trip transaction costs charged on turnover.

4 Results

4.1 Full-sample tearsheet

Table 2 reports full-sample results for the dictionary tier (weekly, 1997–2026, 20bps round-trip TC, $T+1$ execution). FinBERT and fine-tuned-tier results on the full corpus are not reported here — at this project’s measured throughput on CPU (about 8 seconds per document for FinBERT), scoring

the full 3,968-document corpus would take roughly 9 hours; Section 4.3 instead compares all three tiers on a smaller, tractable window.

Strategy	Ann. return	Ann. vol	Sharpe	Max DD	Turnover/yr
Short 10Y	+0.07%	0.92%	0.07	−6.1%	2.2
50/50	−0.24%	0.60%	−0.40	−10.0%	2.9
Short 2Y	−0.55%	0.34%	−1.61	−15.2%	3.6

Table 2: Dictionary-tier full-sample results on real data.

None of the three legs clears a Sharpe ratio of 1 over the full sample; the short-2Y leg is clearly negative. This is a materially weaker result than the same pipeline run on the project’s synthetic data generator (Sharpe up to 1.75), which is expected — the synthetic generator ties tone to returns by construction, real markets do not oblige.

4.2 Sub-period stability

Table 3 breaks the same backtest into five named macro-regimes.

Period	Strategy	Ann. return	Sharpe	Max DD	Hit rate
GFC (2007–2009)	Short 2Y	−0.92%	−2.80	−2.7%	0.25
	Short 10Y	−0.51%	−0.65	−1.8%	0.41
	50/50	−0.71%	−1.35	−2.2%	0.39
ZIRP (2009–2015)	Short 2Y	−0.75%	−4.16	−5.1%	0.21
	Short 10Y	−0.25%	−0.35	−2.3%	0.45
	50/50	−0.50%	−1.18	−3.5%	0.37
COVID (2020–2021)	Short 2Y	−0.44%	−2.90	−1.0%	0.18
	Short 10Y	−0.20%	−0.37	−1.2%	0.40
	50/50	−0.32%	−1.01	−1.1%	0.38
Hiking (2022–2023)	Short 2Y	+0.80%	1.55	−0.4%	0.49
	Short 10Y	+2.37%	1.79	−0.9%	0.54
	50/50	+1.58%	1.83	−0.6%	0.57
Plateau (2024+)	Short 2Y	−0.85%	−2.58	−2.5%	0.27
	Short 10Y	−0.31%	−0.42	−1.9%	0.46
	50/50	−0.58%	−1.12	−2.1%	0.39

Table 3: Dictionary-tier sub-period stability. The signal’s only consistently positive period is the 2022–2023 hiking cycle.

This is one of the paper’s most interesting findings: the full-sample result is negative primarily because of ZIRP, GFC, COVID and the current plateau period, while the 2022–2023 hiking cycle — arguably the regime where central bank tone genuinely diverged most sharply from where it had been anchored for over a decade — is the one period where the signal performs well across every leg. This is consistent with a hypothesis that textual tone carries more information when the policy stance is actually *changing* quickly, and less when it is anchored near a bound (ZIRP) or dominated by an exogenous shock (COVID). We present this as a hypothesis the sub-period table motivates, not as a validated finding — one hiking cycle is one data point.

4.3 Cross-tier comparison

Table 4 compares all three sentiment tiers on a common, tractable sample: a contiguous 2021–2023 window (688 documents — a random subsample across the full history would break the weekly aggregation continuity the signal construction relies on), chosen to include the 2022–2023 hiking cycle identified above plus a year of run-in for the EMA/z-score to stabilise.

Tier	Short 10Y Sharpe	50/50 Sharpe	Short 2Y Sharpe
Dictionary	1.21	0.93	0.02
FinBERT	0.03	−0.34	−1.16
Fine-tuned	−0.63	−1.36	−2.89

Table 4: Sentiment-tier comparison, common 2021–2023 window.

On this window the purpose-built dictionary lexicon outperforms both learned tiers, not the other way around. We read this as plausible rather than surprising: FinBERT is trained on general financial-news sentiment, not central-bank language specifically, and the fine-tuned tier’s training set is small, templated and author-curated (Section 3.1) rather than a large corpus of naturally-occurring, independently-labelled central-bank text. The result is a useful caution against assuming a larger pretrained model automatically outperforms a smaller, domain-specific lexicon on a narrow task with little in-domain labelled data.

5 Limitations

- **No robust full-sample edge.** The headline result is negative-to-flat. We report it because it is true, not because it is the more publishable finding.
- **Small fine-tuned set.** The 482-example fine-tuned tier is author-curated and templated, not independently annotated; its train accuracy (95%) reflects fitting a near-separable templated set, not held-out generalisation to real prose.
- **One hiking cycle.** The sub-period finding rests on a single historical episode (2022–2023); it is suggestive, not statistically established.
- **Duration proxy.** Returns use a constant modified-duration approximation rather than actual bond/future contract specifications; real deployment, particularly on the 2-year leg, would likely use short-term interest rate futures (e.g. SR3) rather than cash bonds, where round-trip costs are closer to 1bp than the 20bps assumed here.
- **Single aggregate rates proxy per currency area.** We do not model the full yield curve or cross-currency spillovers between the three central banks’ communications.

6 Conclusion

A reproducible, real-data pipeline from central bank website to backtest tearsheet does not, on its own, produce a profitable signal with a simple dictionary-tone baseline. It does surface a specific, falsifiable hypothesis — tone-based signals may work better in regime-change periods than in anchored or shock-dominated ones — worth testing against a larger sample of historical hiking/cutting cycles than the single episode available in this dataset. Fully reproducing every

number in this paper takes under two hours end to end: `python -m cb_signal.ingest && python -m cb_signal.run_all --skip-synth --subperiods --walk-forward`.

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